

Latex Showcase (Evolving Document)

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1 Enumerations and Bullets

Often we use enumerations and bullets in our papers. The following shows different ways to use them.

A basic style:

1. First item
2. Second item
3. Third item

This is a enumeration in a single line: (1) aaa, (2) bbb, and (3) ccc.

An enumeration with a custom label (especially useful in presenting research questions):

RQ1 First RQ

RQ2 Second RQ

A basic bullet listing:

- First item
- Second item

A custom bullet listing:

- First item
- Second item

2 Figures

Figure 1 shows an example figure using centering.

Figure 2 shows an example figure using sub-figures side-by-side.

Figure 3 shows another example figure using sub-figures arranged in two rows.

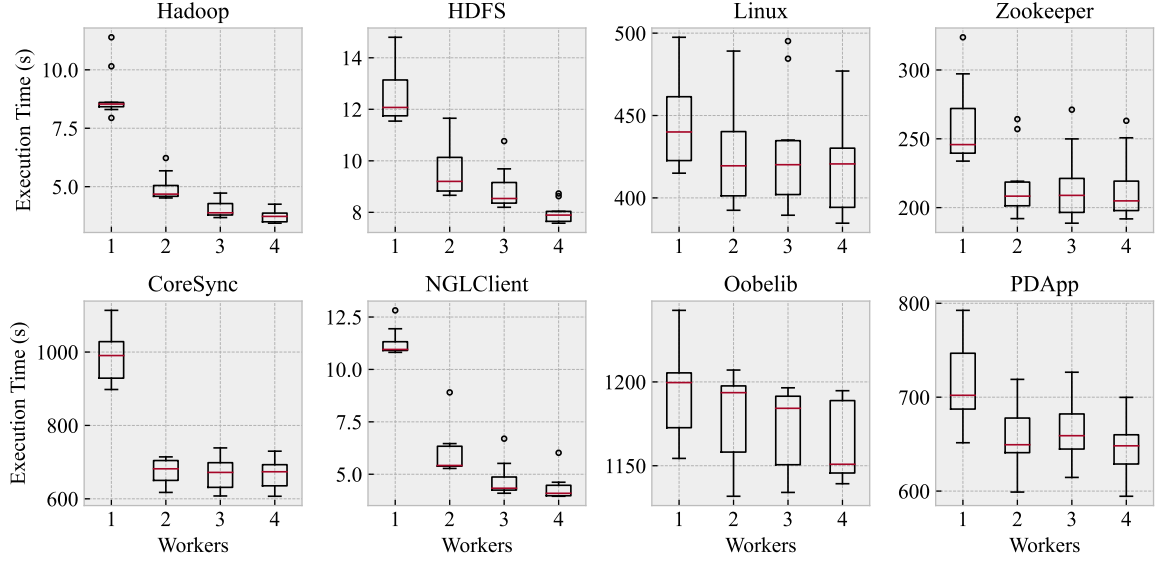
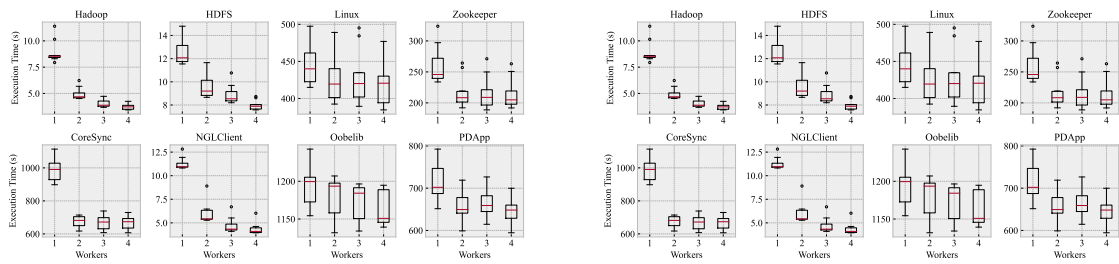


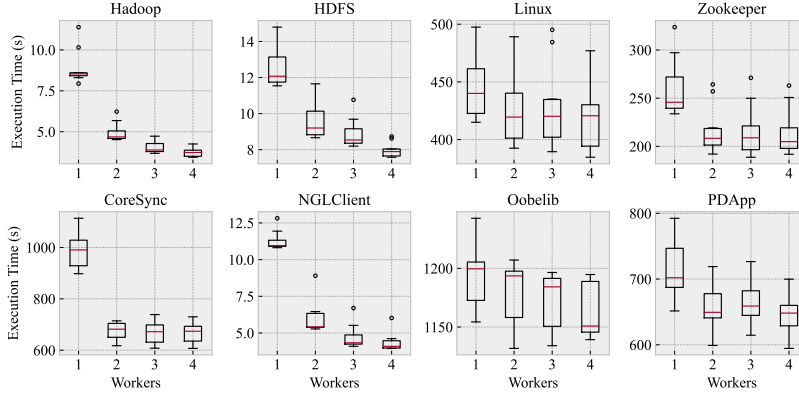
Figure 1: Relationship between the maximum number of parallel workers and the execution time of PRINS [1]



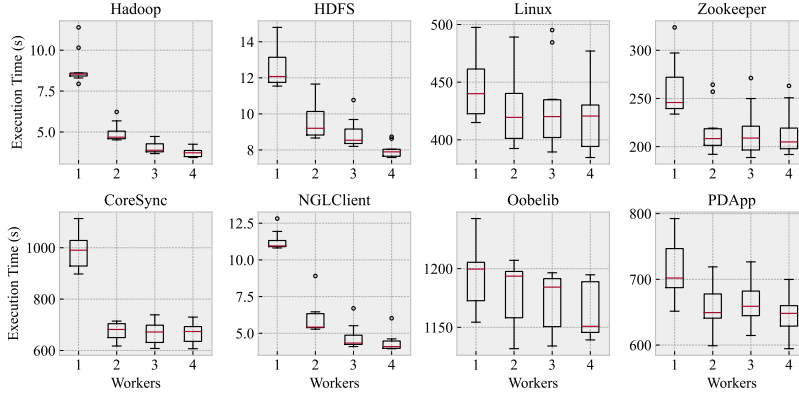
(a) Caption for first sub-figure

(b) Caption for second sub-figure

Figure 2: Caption for the whole figure with sub-figures



(a) Caption for first sub-figure



(b) Caption for second sub-figure

Figure 3: Caption for the whole figure with sub-figures arranged in two rows

Table 1: Subject Systems and Logs [1]

Source	System	# Cmps	# Logs	# Tpls	# Entries	Conf
LogHub	Hadoop	19	68	41	3575	1.00
	HDFS	8	1000	16	18 741	0.95
	Linux	31	42	115	11 259	0.78
	Spark	11	217	21	67 725	0.98
	Zookeeper	18	36	40	25 298	0.91
PC	CoreSync	54	1418	204	30 223	0.94
	NGLClient	27	42	70	892	0.89
	Oobelib	12	250	147	56 557	0.89
	PDApp	10	787	75	47 394	0.87

Table 2: Comparison between MINT (M) and PRINS (P) in terms of Accuracy. Differences between recall (Δ_R), specificity (Δ_S), and balanced accuracy (Δ_B) values are expressed in percentage points (pp); $\mathcal{LDS}$ is the log-component diversity score [1].

System	Recall			Specificity			Balanced Accuracy			
	M	P	Δ_R	M	P	Δ_S	M	P	Δ_B	$\mathcal{LDS}$
Hadoop	1.00	1.00	0.0	0.91	0.88	-3.1	0.96	0.94	-1.5	0.015
HDFS	0.98	0.98	-0.1	0.37	0.72	34.9	0.68	0.85	17.4	0.007
Linux	0.36	0.12	-23.5	0.89	0.98	9.5	0.62	0.55	-7.0	0.561
Zookeeper	0.22	0.10	-11.7	0.93	1.00	7.5	0.57	0.55	-2.1	0.571
CoreSync	0.95	0.93	-1.6	0.85	0.89	4.2	0.90	0.91	1.3	0.048
NGLClient	0.86	0.86	0.0	1.00	0.98	-2.5	0.93	0.92	-1.3	0.195
Oobelib	0.98	0.98	0.0	1.00	1.00	0.0	0.99	0.99	0.0	0.016
PDApp	0.97	0.95	-2.3	0.98	0.98	-0.1	0.97	0.96	-1.2	0.014
Average	0.79	0.74	-4.9	0.87	0.93	6.3	0.83	0.83	0.7	0.178

3 Tables

Table 1 is an example table using an external csv file and multirow. Table 2 is an example table using both an external data, multi-rows, coloured rows, and multi-columns.

4 Listings

Figure 4 shows an example listing using the Python language semantic.

```
(2) db = DB.init(mode="default")
(4) item = getItem(db)
(6) if check(item) is "error":
(7)     logger.error("error in item: %s" % item)
```

Figure 4: Program slice S_r of the program P_{ex} when s_7 and its variable `item` are used as the slicing criterion [2]

5 Algorithms

Algorithm 1: Simple Random Search (for maximisation)

Input : Input Domain X ,

Fitness Function $f : X \rightarrow Y$

Output: Best Solution $x_{best} \in X$ such that $x_{best} \approx \operatorname{argmax}_{x \in X} f(x)$

```
1 Best Solution  $x_{best} \leftarrow$  a candidate solution randomly sampled from  $X$ 
2 while  $x_{best}$  is the ideal solution or we have run out of time do
3   | Candidate Solution  $x_c \leftarrow$  a candidate solution randomly sampled from  $X$ 
4   | if  $f(x_c) > f(x_{best})$  then
5   |   |  $x_{best} \leftarrow x_c$ 
6 return  $x_{best}$ 
```

It is important to use the type information for each variable (e.g., “Candidate Solution” in front of x_c in line 3) since it significantly improves the readability of the algorithm.

6 Definitions and Theorems

Definition 1 (Log Parsing as an Abstraction Function). Given a set of log messages M and a generic set of parsing results A , a log parsing approach can be represented as an abstraction function $\tau: M \rightarrow A$.

See [3] for more examples (and how to write a theoretical paper).

7 Coloured Box

We often need to highlight some text in a coloured box, especially when we want to provide a compact summary of an evaluation result. The following shows how to do it.

The answer to RQ1 is that ...

References

- [1] D. Shin, D. Bianculli, and L. Briand, “PRINS: scalable model inference for component-based system logs,” *Empirical Software Engineering*, vol. 27, no. 4, p. 87, 2022.
- [2] J. H. Dawes, D. Shin, and D. Bianculli, “Towards log slicing,” in *International Conference on Fundamental Approaches to Software Engineering*. Springer Nature Switzerland Cham, 2023, pp. 249–259.

- [3] D. Shin, Z. A. Khan, D. Bianculli, and L. Briand, “A theoretical framework for understanding the relationship between log parsing and anomaly detection,” in *Runtime Verification*, L. Feng and D. Fisman, Eds. Cham: Springer International Publishing, 2021, pp. 277–287.